

Component	Configuration
<i>Aerial platform and autonomy stack</i>	
Airframe	in-house quadrotor, about 0.8 m rotor span, roll cage
Ranging sensor	Ouster spinning 3D lidar (360° azimuth)
Inertial sensing	VectorNav IMU
Onboard computer	Intel NUC-class single-board computer (full autonomy stack)
Flight controller	Pixhawk Cube Orange (attitude stabilisation)
Power	lithium high-voltage (LiHV) battery
Context camera	first-person-view camera, separately powered, not used by autonomy
State estimation	DLIO lidar-inertial odometry, onboard, no external fix
Mapping	hashed TSDF/ESDF volumetric map, 0.25 m voxels, 15 m ray integration
Planning	STAGE graph-based exploration (Supplementary Methods)
Supervision	behaviour tree (Extended Data Fig. 2b), onboard
<i>Surface photogrammetric survey</i>	
Aircraft, camera	senseFly eBee X RTK fixed wing, S.O.D.A. camera (10.6 mm, 5472 × 3648 px)
Acquisition	28 August 2025, 135 m above ground, 964 images, 2.08 km ² , no snow
Processing	Agisoft Metashape Professional 2.1.3; 7.43 million tie points, 0.48 px reprojection error
Products	dense cloud 6.99 × 10 ⁸ points (3.08 cm ground sampling distance); DEM at 15.6 points m ⁻² ; orthomosaic
Georeferencing	onboard GNSS with IceCORS real-time correction, camera stations to 9.5 cm; three ground control points (Trimble R8s), 14.0 cm total RMSE, 7.6 cm in height
Reference frame	ISN2016 / Lambert 2016, orthometric heights (ISH2004 geoid); delivered ellipsoidal heights differ by 66.00 m at the site

Parameter	Symbol	Value
Vehicle bounding box		1.2 × 1.2 × 1.2 m
Clearance inflation margin	δ	0.4 m
Path safety extension		1.5 m
Gain-model sensor range	$r_{\max}$	8 m
Gain-model ray spacing		0.12 rad
Local sampling space		±15 m cube
Maximum local graph size		700 vertices
Maximum edge length		2.0 m
Executed path truncation		8 m
Gain weight (applied to the normalised gain $\hat{g}$)	w_g	60
Heading-change penalty weight	w_ψ	0.1
Path-length penalty weight	w_d	0.04
Cruise / homing speed limit		2.0 / 0.9 m s ⁻¹
Yaw rate limit		0.15 rad s ⁻¹
Exploration time budget		1,800 s
Onboard map voxel size	ℓ	0.25 m
Map ray integration length	r_{map}	15 m
ESDF truncation distance		2.0 m